

The Method of Characteristics for a General Quasi-linear First-Order PDE :

$$a(x,y,u) \frac{\partial u}{\partial x} + b(x,y,u) \frac{\partial u}{\partial y} = c(x,y,u) \quad \text{---} *$$

Note that every linear first-order PDE is also a semi-linear first-order PDE; and every semi-linear first-order PDE is also a quasi-linear first-order PDE: $\text{Linear} \Rightarrow \text{Semi-linear} \Rightarrow \text{Quasi-linear}$.

$$a(x,y)u_x + b(x,y)u_y + c(x,y)u = f(x,y) \leftarrow \text{Linear}$$

$$a(x,y)u_x + b(x,y)u_y = f(x,y,u) \leftarrow \text{Semi-linear}$$

$$a(x,y,u)u_x + b(x,y,u)u_y = f(x,y,u) \leftarrow \text{Quasi-linear}$$

Lagrange-Charpit Equations :

$$\boxed{\frac{dx}{a} = \frac{dy}{b} = \frac{du}{c}} \quad \text{---} **,$$

$$\begin{cases} \frac{du}{dx} = \frac{c(x,y,u)}{a(x,y,u)} \text{ and } \frac{dy}{dx} = \frac{b(x,y,u)}{a(x,y,u)} \\ \text{or} \\ \frac{du}{dy} = \frac{c(x,y,u)}{b(x,y,u)} \text{ and } \frac{dx}{dy} = \frac{a(x,y,u)}{b(x,y,u)} \end{cases}$$

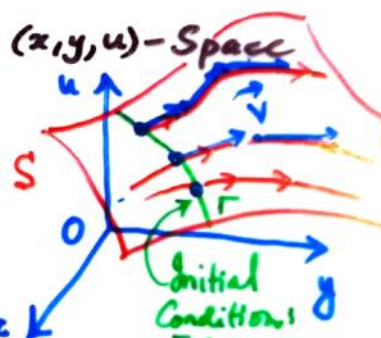
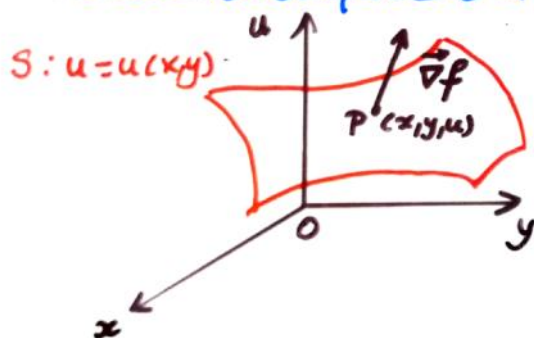
The above pair of equations, $(*)$, are called the **Characteristic Equations** of the quasi-linear PDE, $(*)$.

The Equation: $\boxed{\frac{dy}{dx} = \frac{b(x,y,u)}{a(x,y,u)}}$

A family of curves in (x,y,u) -space that sit in the solution surface S — these curves are called the **Characteristics** of the PDE. (after Cauchy).

Comments:

Assume the solution of $(*)$ in the form: $u = u(x,y)$ — it represents the solution surface S in $\mathbb{R}^3$.



$S: u = u(x,y)$ (explicit form) $\longrightarrow$ $f(x,y,u) = u(x,y) - u = 0$. (implicit form)

- 1) The projection of a characteristic curve in the (x,y) -plane ($u=0$) is called a **characteristic base curve**, a **projected characteristic curve**, or simply **characteristic**.
- 2) $(*)$ has only two independent ODEs. Hence, it is a two-parameter family of characteristic curves in (x,y,u) -space: $F(x,y,u,A,B)=0$ and $G(x,y,u,A,B)=0$.

Ex: Find the general solution of quasi-linear PDE:

*Linear
Homogeneous → First-Order
Constant Coefficients* $a \frac{\partial u}{\partial x} + \frac{\partial u}{\partial t} = 0.$

Our $b=1$ and $c=0$. The reciprocal Lagrange-Charpit Equations:

$$\frac{a}{dx} = \frac{1}{dt} = \frac{0}{du}$$

$$\therefore \frac{dt}{dx} = \frac{1}{a} \text{ or } \frac{dx}{dt} = a \Rightarrow x = at + A, \quad \begin{array}{l} A \text{ arbitrary} \\ \text{constant.} \end{array}$$

$$\frac{du}{dt} = 0 \Rightarrow u = \text{constant} = B, \quad \begin{array}{l} B \text{ is an} \\ \text{arbitrary constant.} \end{array}$$

$\therefore \begin{array}{l} x = at + A \\ u = B \end{array} \left\} \begin{array}{l} \text{a two-parameter} \\ \text{family of characteristic} \\ \text{curves.} \end{array} \right.$

Specifying B as a function of A defines a one-parameter family of characteristic curves on a solution surface.

The general solution: $B = h(A).$

$\therefore u(x,y) = h(x-at)$, where h is an arbitrary function.

Ex: Check $u(x,y) = h(x-at)$ satisfies the PDE.

$\therefore u(x,t) = h(A) = h(x-at).$ 

Ex: Find the general solution of the quasi-linear PDE:

*Linear
Non-homogeneous
First-Order
Variable Coefficients* $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = u$

Note: Our $a=x$, $b=y$, $c=u$.

The Lagrange-Charpit Equations:

$$\frac{dx}{x} = \frac{dy}{y} = \frac{du}{u}$$

Hence,

Separation of Variables:
 $\frac{dx}{dy} = \frac{x}{y} \Rightarrow \ln x = \ln y + \ln A$

$\Rightarrow x = Ay$, A arbitrary constant.

Separation of Variables:
 $\frac{du}{dy} = \frac{u}{y} \Rightarrow \ln u = \ln y + \ln B$

$\Rightarrow u = yB$.

$\therefore \frac{x}{y} = A$ and $\frac{u}{y} = B$ ← Two-Parameter family of Characteristic Curves

Specifying B as a function of A defines one-parameter family of characteristic curves: $B = h(A)$, a solution surface.

The general solution $B = h(A)$.

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$\therefore u(x, y) = y h\left(\frac{x}{y}\right)$, h an arbitrary function.

NOTE: Had we selected the equations of the form:

$$\frac{dy}{dx} = \frac{y}{x} \text{ and } \frac{du}{dx} = \frac{u}{x}.$$

The general solution: $u(x, y) = x g\left(\frac{y}{x}\right)$, g is an arbitrary function.

Ex: Find the general solution of the PDE: $u_x + y u_y = 0$.

The Lagrange-Charpit Equations:

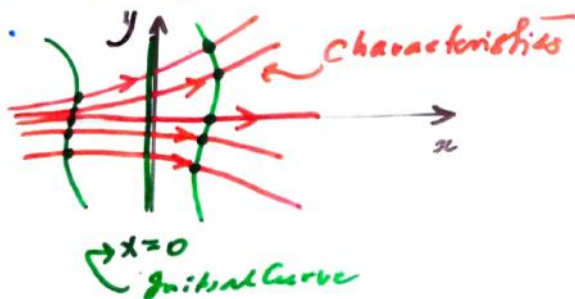
etc: $a = 1, b = y, c = 0$.

$$\frac{1}{dx} = \frac{y}{dy} = \frac{0}{du} \Rightarrow \frac{dy}{dx} = y \text{ and } \frac{du}{dx} = 0.$$

$$\therefore y = Ae^x \text{ and } u = \text{const} = B.$$

The general solution: $B = h(A)$.

$u(x,y) = h(e^{-x}y)$, where h is an arbitrary function.



Ex: Find the solution of PDE: $u_x + yu_y = 0$ which satisfies the auxiliary condition $u(0,y) = y^3$.

We know the general solution: $u(x,y) = h(e^{-x}y)$, where h is an arbitrary function.

We use the auxiliary condition to fix the function h .

$$u(0,y) = y^3 \Rightarrow h(e^{-0}y) = y^3$$

$$\Rightarrow h(y) = y^3. \quad h(x) = x^3$$

$$\therefore \text{our particular solution: } u(x,y) = (e^{-x}y)^3 = e^{-3x}y^3.$$

$$\therefore u(x,y) = y^3 e^{-3x}.$$

Ex: Solve the PDE: $u_x + 2xy^2u_y = 0$ subject to the 67 initial condition $u(0,y) = y^3$.

We know that the general solution is $u(x,y) = h(x^2 + \frac{1}{y})$, where h is an arbitrary function.

$$u(0,y) = y^3 \Rightarrow h(\frac{1}{y}) = y^3$$

$$\Rightarrow h(s) = \frac{1}{s^3} \quad (\text{Put } s = \frac{1}{y}.)$$

∴ Our particular solution:

$$u(x,y) = \frac{1}{(x^2 + \frac{1}{y})^3} = (x^2 + \frac{1}{y})^{-3}$$

Ex: Solve the PDE: $u_x + 2xy^2u_y = 0$.

The Lagrange - Charpit Equations:

$$\frac{1}{dx} = \frac{2xy^2}{dy} = \frac{0}{du}$$

Ex*: Find the general solution of quasi-linear PDE
$$x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} = (x+y)u$$

Our equation is of the form: $a \frac{\partial u}{\partial x} + b \frac{\partial u}{\partial y} = c$

$$\therefore a = x^2, b = y^2, c = (x+y)u.$$

$$\therefore \frac{dx}{x^2} = \frac{dy}{y^2} = \frac{du}{(x+y)u}$$

$$\Rightarrow \frac{dx}{dy} = \frac{x^2}{y^2} \xrightarrow{\text{Separation of Variables}} \frac{dx}{x^2} = \frac{dy}{y^2} \Rightarrow \bar{x}' = \bar{y}' + A,$$

where A an arbitrary constant: Hence, $\frac{xy}{x+y} = A$ — (1)

Note: $dx = \frac{x^2}{(x+y)u} du$ and $dy = \frac{y^2}{(x+y)u} du$.

Subtracting:

$$dx - dy = \frac{x^2 - y^2}{(x+y)u} du$$

$$\Rightarrow \frac{dx - dy}{x - y} = \frac{du}{u} \quad \frac{x-y}{u} = B$$

$$\Rightarrow \frac{d(x-y)}{x-y} = \frac{du}{u} \quad \frac{u}{x-y} = \frac{1}{B}$$

$$\Rightarrow \ln(x-y) = \ln u + \tilde{B}$$

$$\Rightarrow \frac{u}{x-y} = B \quad \text{--- } x^y,$$

3 specifying B as a function of $A: B = f(A) \leftarrow$ The General Solution

$\therefore u(x,y) = (x-y)f\left(\frac{xy}{x+y}\right)$, where f is an arbitrary function.

Ex: Find the general solution of the PDE: $2y \frac{\partial u}{\partial x} + u \frac{\partial u}{\partial y} = 2yu^2$.

Ex: Solve the Cauchy problem $xu_x + yu_y = xe^{-u}$ subject to $u(x,y) = x$ when $y = x^2$.

Ex: Find the solution to the initial value problem (IVP):

$$u_t + u u_x = 0, \quad x \in \mathbb{R}, t > 0;$$

$$u(x,0) = f(x), \quad \forall x \in \mathbb{R}.$$

Ex: Find the solution to the initial value problem (IVP):

$$\begin{cases} u_t + u_x = u, & x \in \mathbb{R}, t > 0; \\ u(x,0) = f(x), & \forall x \in \mathbb{R}. \end{cases}$$